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Passive acoustic mapping utilizing optimal beamforming in ultrasound therapy monitoring

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Passive acoustic mapping (PAM) is a promising imaging method that enables real-time three-dimensional monitoring of ultrasound therapy through the reconstruction of acoustic emissions passively received on an array of ultrasonic sensors. A passive beamforming method is presented that provides greatly improved spatial accuracy over the conventionally used time exposure acoustics (TEA) PAM reconstruction algorithm. Both the Capon beamformer and the robust Capon beamformer (RCB) for PAM are suggested as methods to reduce interference artifacts and improve resolution, which has been one of the experimental issues previously observed with TEA. Simulation results that replicate the experimental artifacts are shown to suggest that bubble interactions are the chief cause. Analysis is provided to show that these multiple bubble artifacts are generally not reduced by TEA, while Capon-based methods are able to reduce the artifacts. This is followed by experimental results from *in vitro* experiments and *in vivo* oncolytic viral therapy trials that show improved results in PAM, where RCB is able to more accurately localize the acoustic activity than TEA. © 2015 Acoustical Society of America.

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I. INTRODUCTION

Therapeutic ultrasound offers tremendous promise as a tool for non-invasive surgery or enhanced and targeted drug delivery, for applications ranging from deep-seated tumor ablation by high intensity focused ultrasound (HIFU) to sonothrombolysis for stroke therapy. The key underlying mechanisms are either thermal, arising from viscous absorption of the therapeutic pulse, or mechanical, associated with sustained or unstable acoustically driven oscillations of microbubbles known as stable and inertial cavitation.¹ Both for surgical and drug delivery applications, the ability to monitor and map the extent of treatment in real time is crucial to therapeutic outcomes. Presently available treatment monitoring tools include conventional B-mode ultrasound (US), which is cost-effective but not capable of mapping sub-boiling heating or inertial cavitation, and magnetic resonance imaging (MRI), which is very-high-cost and can only directly map tissue heating at modest frame rates (~ 1 Hz).

Passive acoustic mapping (PAM) is a novel US-based technique² that enables real-time three-dimensional mapping of sources of nonlinear acoustic emissions, including stable

and inertial cavitation of bubbles induced by an incident US field, by using the signals passively received on a conventional or customized US imaging array during therapy. PAM has been shown to allow successful monitoring of lesion formation by HIFU in tissue³ through simultaneous mapping of harmonic emissions arising from nonlinear scattering of the incident therapeutic pulse, and broadband emissions arising from bubble collapses (inertial cavitation) caused by incident US narrowband exposure. More recently, PAM has also been proposed as a tool for real-time mapping of drug delivery to tumors⁴ by enabling spatiotemporal monitoring of inertial cavitation processes responsible for drug release from carriers or enhanced extravasation of free agents, such as oncolytic viruses.⁵ PAM has also been used as a means for monitoring therapies in the brain,⁶ with added skull corrections suggested to improve images.⁷ PAM or variants thereof^{8,9} constitute an active area of research in US therapy because of the need of therapy monitoring; furthermore, PAM enables one modality to guide, deliver, and monitor treatment, which makes US therapy an attractive and cost-effective treatment option.

There are however several challenges to passively mapping acoustic emissions during ultrasound therapy in a clinical setting. Depending on the nature of the cavitation activity and other acoustic sources being mapped, the signals

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received can have very different frequency-domain signatures, and could either be broadband or narrowband in nature. For reasons of cost and simplicity, it is also important to achieve adequate spatial resolution with the minimum number of channels (typically in the range 64–128), prompting the recent exploration of sparse array techniques.¹⁰

Most importantly, however, the choice of reconstruction algorithm used to generate maps from channel data of acoustic emissions must be able to deal with both the complex propagation medium and scattering of the body. Previous work describing PAM (Ref. 2) has been conducted with the time exposure acoustics (TEA) algorithm¹¹ [hereafter termed time exposure acoustics-passive acoustic mapping (TEA-PAM)], which is most closely related to delay and sum (DAS) beamforming. Using such a basic beamforming algorithm with wideband data in the near-field, the resulting images have relatively coarse resolution as, in clinically-relevant configurations, TEA-PAM is only able to achieve an axial resolution determined by the physical aperture and the signal bandwidth.¹² Furthermore, TEA-PAM is subject to imaging artifacts arising from variability of the sensors (gain, aberrations, incorrect positions) in the array and propagation through non-homogeneous media, as well as interference from scatterers that are located around the focal point. Even though the aperture and bandwidth of the receiving array will typically be determined by the geometric requirements of a specific clinical application, it is suggested that some improvement to PAM can be achieved by enhanced algorithms. For example, there have been many significant advances in data adaptive algorithms in radar, sonar, and communications research over the last 30 years that may be suitably modified to the PAM problem.

Optimal adaptive beamforming using constraints has recently been introduced in several biomedical imaging applications as a means of overcoming the limitations of DAS. In active US imaging, researchers have used Capon's algorithm¹³ or minimum variance distortionless response (MVDR)^{14–16} to improve B-mode imaging. A variant on Capon's algorithm, the robust Capon beamformer (RCB),^{17,18} is of special interest as it is known to have better resolution and interference suppression than DAS, and has been shown with active US imaging.¹⁹ In addition, it can deal with array calibration uncertainty, a concern when using commercial diagnostic US probes as the passive receiving array; and steering vector mismatch, which is an issue with non-homogeneous media like the human body. The RCB has further been used in other applications like breast cancer detection in thermal acoustic imaging²⁰ and microwave imaging.²¹

In this paper, the RCB is instead modified to deal with the passive mapping of acoustic cavitation of collapsing bubbles during ultrasound therapy (UT) monitoring in the context of drug delivery, and is thereby termed robust Capon beamformer-passive acoustic mapping (RCB-PAM). RCB-PAM is shown to localize cavitation activity and remove imaging artifacts in simulated bubble data, where TEA-PAM had poor performance. In both *in vitro* and *in vivo* experimental conditions, RCB-PAM was shown to reduce the region of activity to only areas where, by design, cavitation

was expected to occur, and additionally it was able to operate with poorly calibrated arrays, which in both cases TEA-PAM was unable to match. The paper is organized into five sections. Section II provides background details on UT monitoring using PAM, the TEA, and RCB algorithms, and results that demonstrate the improvements obtained by optimal beamforming. The simulations, *in vitro*, and *in vivo* drug delivery experiments that are used to compare optimal beamforming to TEA are also described in Sec. II. Section III presents results of both simulations and experiments, and Sec. IV contains a summary of all findings.

II. METHODS

A. Algorithm: PAM background

In active, B-mode US applications the goal is to image the intensity of permanent (though potentially moving) linear and non-linear scatterers of the incident sound field, where knowledge of the timing of a transmitted pulse is a key parameter. In PAM,^{22–24} the goal is instead to image the broadband acoustic emissions, without requiring any knowledge of the timing or nature of the physical excitation which has excited such sources. Such nonlinear sources include inertially collapsing bubbles, which produce broadband acoustic emissions, or stably oscillating bubbles, which typically produce narrowband harmonic emissions. While these bubbles are, in the present work, excited by a narrowband incident US field, PAM still works without alteration whether the bubble activity was caused by US, hydrodynamic cavitation, or the photoacoustic effect.

Consider the experimental setup shown in Fig. 1(a), where two identical, geometrically-focused HIFU transducers are co-aligned to produce a small focal region. Positioned between the transducers and co-aligned to the same focal region is a diagnostic, linear array, US imaging probe.²⁵ The US imaging probe, made up of $j = 1, \dots, N$ elements as shown in Fig. 1(b), is used to passively record the acoustic signals generated during US exposure generated by the HIFU transducer. The orientation of the diagnostic array is centered on the HIFU focal area, from where the acoustic signals will predominantly emanate. The dual-transducer setup was designed to create a very small focal volume with reasonably sized transducers for use in *in vivo* animal experiments.

Once the data are collected, the first step in the PAM process, whether TEA or RCB is used, is to create a grid of voxel locations over which the image is computed with some resolution. Let the voxel grid be defined by point locations in the set B , where each point in B is a three-element vector $\mathbf{x} = [x, y, z]^T$, with voxel grid spacing in each dimension given by $[\delta_x, \delta_y, \delta_z]^T$ (superscript T denotes transpose). Generally, the imaging region defined by the set B is the area surrounding the HIFU focus, and can be any pairwise combination of the transverse, elevation, and axial planes, or all three planes for full volumetric imaging. Then for each voxel, the data is pre-steered with time delays applied to each sensor signal to compensate for the propagation time from the voxel being imaged to the sensors.

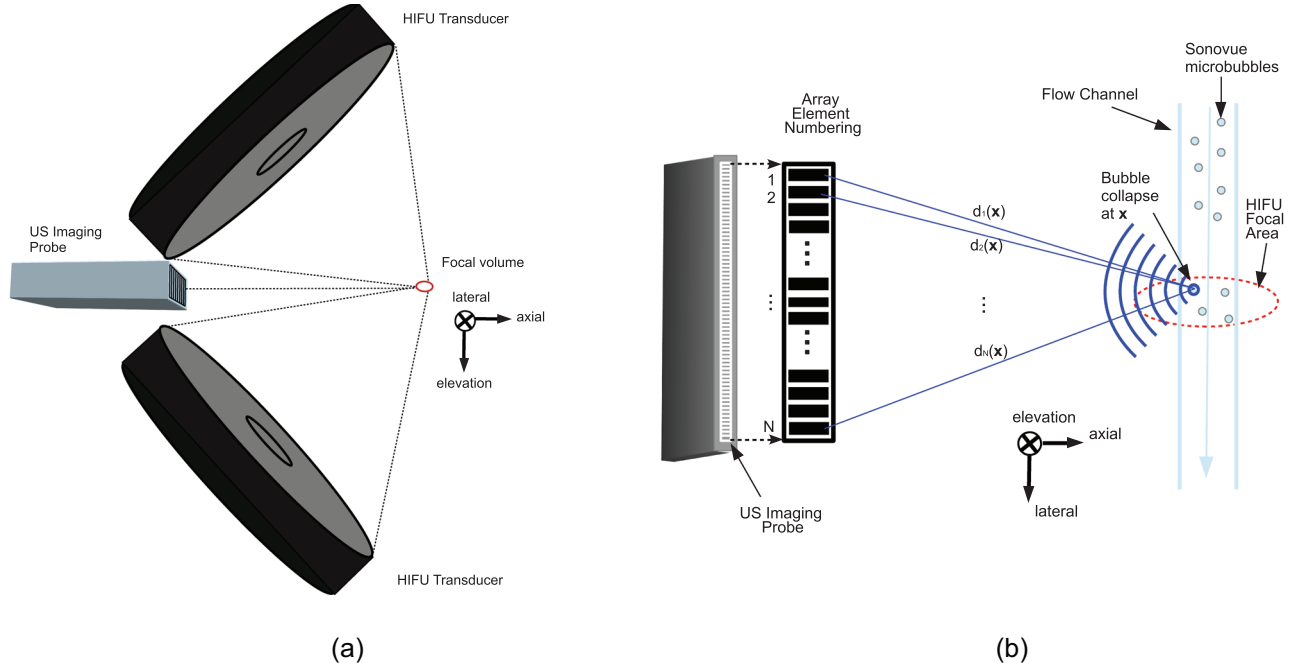


FIG. 1. (Color online) Dual HIFU transducer and diagnostic imaging array geometry (a) for *in vitro* and *in vivo* experiments and (b) detail of the array geometry relative to the HIFU focus for the case of the *in vitro* flow channel experiment. Note that in the *in vivo* animal experiment, the flow channel is replaced and the HIFU focus is positioned on the tumor.

First, we define the distance from voxel position $\mathbf{x}$ to sensor j at position $\mathbf{x}_j$ by

$$d_j(\mathbf{x}) = \sqrt{(x - x_j)^2 + (y - y_j)^2 + (z - z_j)^2}. \quad (1)$$

Then the data can be steered to this position, and the source strength measured by the array is

$$q(\mathbf{x}, t) = \frac{4\pi}{N\alpha} \sum_{j=1}^N d_j(\mathbf{x}) s_j(t + \tau_j(\mathbf{x})), \quad (2)$$

where $s_j(t)$ is the recorded signal at sensor j , $\tau_j(\mathbf{x}) = d_j(\mathbf{x})/c$ is the propagation time from location $\mathbf{x}$ to sensor j , α is the piezoelectric coefficient that is a characteristic of each sensor element, and c is the speed of propagation. The multiplication by $d_j(\mathbf{x})$ in Eq. (2) compensates for the reduction in source signal strength from spherical spreading of the wavefronts.

In TEA-PAM, an image of source energy is generated by integrating the square of the source strength over a collection period² of length T , assuming uniform array weighting,

$$\Psi(\mathbf{x}) = \frac{1}{4\pi\rho_o c} \int_{t_0}^{t_0+T} q(\mathbf{x}, t)^2 dt, \quad (3)$$

where ρ_o is the density of the medium and c is the speed of propagation. The energy $\psi(\mathbf{x})$ is computed for each voxel position $\mathbf{x} \in B$ in the grid, creating the entire map of the acoustic activity.

To aid in later developments in comparing TEA to optimal beamforming methods, we expand the previous PAM terminology with a complete signal model. We assume that the signal measured at each sensor is a superposition of K

source signals (bubbles), with delays determined by the propagation time from each source to each sensor, and additive noise $v_j(t)$,

$$s_j(t) = \sum_{k=1}^K \frac{1}{d_j(\mathbf{x}_k)} p_k(t - \tau_{j,k}) + v_j(t), \quad j = 1, \dots, N, \quad (4)$$

where $\tau_{j,k} = d_j(\mathbf{x}_k)/c$ is the propagation time from source k to sensor j , the division by $d_j(\mathbf{x}_k)$ accounts for the source signal strength reduction from spherical spreading, and $p_k(t)$ is the signal waveform emitted from source k that would be measured at one radial unit from the position of source k . It should be noted that the imaging algorithms are, generally speaking, independent of the form of the bubble signal waveform $p_k(t)$, though correlation between sources is covered in Sec. III A. The additive noise is assumed to be temporally white and independent at each sensor with zero mean. The first step, as above for TEA, is to create the voxel grid and pre-steer the data for each position in the grid. This is written explicitly as

$$s_j(t + \tau_j(\mathbf{x})) = \sum_{k=1}^K \frac{1}{d_j(\mathbf{x}_k)} p_k(t + \tau_j(\mathbf{x}) - \tau_{j,k}) + v_j(t + \tau_j(\mathbf{x})), \quad j = 1, \dots, N, \quad (5)$$

where $\tau_j(\mathbf{x}) = d_j(\mathbf{x})/c$ is the delay on sensor j to pre-steer to imaging location $\mathbf{x}$. If the pre-steered data at the N sensors is defined as the vector

$$\mathbf{s}(\mathbf{x}, t) = \begin{bmatrix} s_1(t + \tau_1(\mathbf{x})) \\ \vdots \\ s_N(t + \tau_N(\mathbf{x})) \end{bmatrix}, \quad (6)$$

then the correlation matrix can be expressed as follows for a measurement interval of length T :

$$\mathbf{R}_s(\mathbf{x}) = \int_{t_0}^{t_0+T} \mathbf{s}(\mathbf{x}, t) \mathbf{s}(\mathbf{x}, t)^T dt. \quad (7)$$

The expression in Eq. (2) for TEA-PAM may be generalized to a beamforming operation in which an $N \times 1$ weight vector $\mathbf{w}$ is applied to the sensor data after multiplying each sensor signal by $d_j(\mathbf{x})$ to account for the spherical spreading. We define a diagonal matrix $\mathbf{D}$ of distances from each sensor to the voxel $\mathbf{x}$ of interest,

$$\mathbf{D}(\mathbf{x}) = \text{diag}[d_1(\mathbf{x}), \dots, d_N(\mathbf{x})]. \quad (8)$$

Then the general expression for the output time signal with beamforming weights $\mathbf{w}$ and scaling by $4\pi/\alpha$ as in Eq. (2) is the linear combination

$$\begin{aligned} q(\mathbf{x}, t) &= \frac{4\pi}{\alpha} \mathbf{w}^T \mathbf{D}(\mathbf{x}) \mathbf{s}(\mathbf{x}, t) \\ &= \frac{4\pi}{\alpha} \sum_{j=1}^N w_j d_j(\mathbf{x}) s_j(t + \tau_j(\mathbf{x})), \end{aligned} \quad (9)$$

and the source energy at imaging location $\mathbf{x}$ in Eq. (3) can be expressed as

$$\begin{aligned} \Psi(\mathbf{x}) &= \frac{1}{4\pi\rho_o c} \int_{t_0}^{t_0+T} q(\mathbf{x}, t)^2 dt \\ &= \frac{4\pi}{\alpha^2 \rho_o c} \mathbf{w}^T \mathbf{D}(\mathbf{x}) \mathbf{R}_s(\mathbf{x}) \mathbf{D}(\mathbf{x}) \mathbf{w}. \end{aligned} \quad (10)$$

In TEA-PAM, the weight vector is uniform across the array, i.e., $\mathbf{w}_{\text{TEA}} = (1/N)\mathbf{1}$, where $\mathbf{1}$ is a $N \times 1$ vector of ones. Note that setting $\mathbf{w} = \mathbf{w}_{\text{TEA}}$ in Eqs. (9) and (10) yields Eqs. (2) and (3). In TEA-PAM, the beamforming weights are not only uniform, but fixed for all voxels. If instead the weights are chosen for each voxel in the image in a data-adaptive way according to optimality criteria, then the effect of interference from other source locations may be significantly reduced.

To aid in analysis in Sec. III A, note that for many imaging scenarios in this paper, the distances $d_j(\mathbf{x})$ in the matrix $\mathbf{D}$ in Eq. (8) are all nearly equal to the mean distance

$$d(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^N d_j(\mathbf{x}), \quad (11)$$

so $\mathbf{D}$ is nearly a scalar diagonal matrix,

$$\mathbf{D}(\mathbf{x}) \approx d(\mathbf{x}) \mathbf{I}. \quad (12)$$

The approximation in Eq. (12) yields the following TEA-PAM approximation expressions for Eqs. (9) and (10):

$$q(\mathbf{x}, t) \approx \frac{4\pi d(\mathbf{x})}{\alpha} \mathbf{w}^T \mathbf{s}(\mathbf{x}, t), \quad (13)$$

$$\Psi(\mathbf{x}) \approx \frac{4\pi d(\mathbf{x})^2}{\alpha^2 \rho_o c} \mathbf{w}^T \mathbf{R}_s(\mathbf{x}) \mathbf{w}, \quad (14)$$

which we will use later to compare the characteristics of TEA-PAM with RCB-PAM. This approximation is only used for analysis in Sec. III A, and not in the algorithms used to actually process data.

B. Algorithm: Optimal beamforming for improved PAM

One of the key challenges of PAM for monitoring of UT is correlating the acoustic activity observed in the PAM image to the location of the bioeffect in tissue observed post treatment.²⁶ This could mean correlating PAM to a lesion demonstrating thermal necrosis for ablation-based UT, or in the case of drug delivery, correlating drug release as verified by cytotoxic necrosis. For PAM to truly benefit the clinician, the region of cavitation activity must be accurately localized with the **highest possible resolution and free from algorithmic artifacts**. In both *in vitro* and *in vivo* experiments, an effect occurs in the PAM images using TEA whereby when ensonifying at even modest peak negative pressures (<1 MPa), the acoustic region of activity appears broader and more elongated axially than would be expected, and this effect can be seen in the left panels of Fig. 2 (*in vitro*) and Fig. 3 (*in vivo*).

Optimal beamforming algorithms have been used extensively in application areas such as radar, sonar, communications, and others²⁷ to suppress interference and improve the resolution of DAS beamforming. Several methods have been developed that use the measured data as a means of adapting a weight vector to the data through optimization of some criterion. One such method, the Capon beamformer (CB),¹³ also known as the MVDR beamformer,²⁸ is a data-adaptive method that can be used to compute the weight vector $\mathbf{w}$ based on the pre-steered and amplitude-compensated data vector $\mathbf{D}(\mathbf{x}) \mathbf{s}(\mathbf{x}, t)$ for a particular imaging location $\mathbf{x}$. The weight vector depends on $\mathbf{x}$, but the functional dependence of $\mathbf{w}$ on $\mathbf{x}$ is suppressed for simplicity. In array processing terminology, the quiescent array steering vector, describes in vector form the path from source to sensor, and would be of the form $\mathbf{a}_0 = [d_1^{-1}(\mathbf{x}_k), \dots, d_N^{-1}(\mathbf{x}_k)]^T$ using previous notation and assuming perfect array calibration (i.e., exact array element positions, gains, phases, and no cross-coupling). However, for received data already pre-steered and distance compensated as described by the vector $\mathbf{D}(\mathbf{x}) \mathbf{s}(\mathbf{x}, t)$, then the steering vector is simply the unity vector $\mathbf{a}_0 = \mathbf{1}$. The Capon weight vector $\mathbf{w}_{\text{CB}}$ is the solution to the following constrained optimization problem:

$$\min_{\mathbf{w}} \mathbf{w}^T \mathbf{D}(\mathbf{x}) \mathbf{R}_s(\mathbf{x}) \mathbf{D}(\mathbf{x}) \mathbf{w} \text{ subject to } \mathbf{w}^T \mathbf{a}_0 = 1. \quad (15)$$

This beamformer is designed to pass signals undisturbed from the voxel position of interest $\mathbf{x}$, and reject interference from other directions. The solution to Eq. (15) is

$$\mathbf{w}_{\text{CB}} = \frac{\mathbf{D}(\mathbf{x})^{-1} \mathbf{R}_s(\mathbf{x})^{-1} \mathbf{D}(\mathbf{x})^{-1} \mathbf{a}_0}{\mathbf{a}_0^T \mathbf{D}(\mathbf{x})^{-1} \mathbf{R}_s(\mathbf{x})^{-1} \mathbf{D}(\mathbf{x})^{-1} \mathbf{a}_0} \quad (16)$$

and the Capon-based PAM image from Eq. (10) is

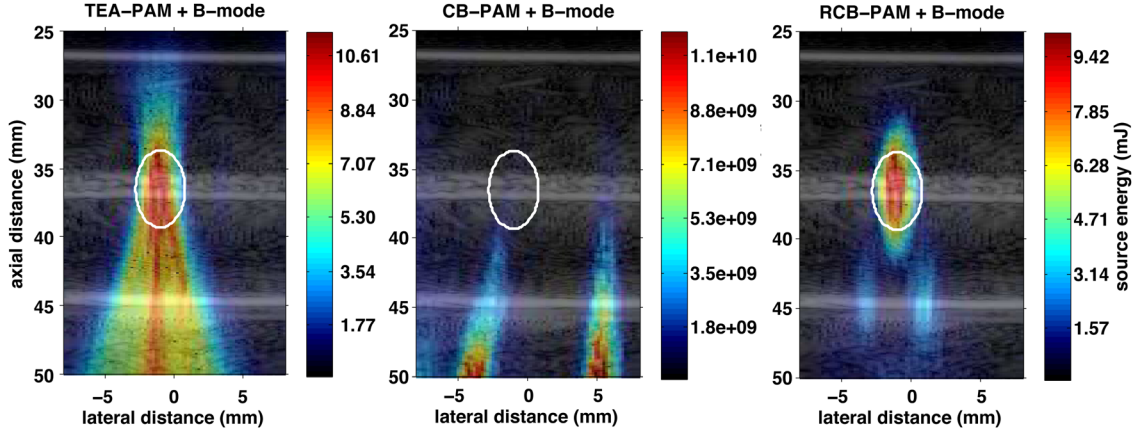


FIG. 2. (Color online) *In vitro* experimental PAM images for TEA-PAM, CB-PAM, and RCB-PAM overlaid on B-mode images for 300 kPa insonification of a tissue phantom with a center flow channel containing SonoVue microbubbles.

$$\Psi_{CB}(\mathbf{x}) = \frac{4\pi}{\alpha^2 \rho_o c} \cdot \frac{1}{\mathbf{a}_0^T \mathbf{D}(\mathbf{x})^{-1} \mathbf{R}_s(\mathbf{x})^{-1} \mathbf{D}(\mathbf{x})^{-1} \mathbf{a}_0}. \quad (17)$$

The approximations to Eqs. (16) and (17) using Eq. (12) are

$$\mathbf{w}_{CB} \approx \frac{\mathbf{R}_s(\mathbf{x})^{-1} \mathbf{a}_0}{\mathbf{a}_0^T \mathbf{R}_s(\mathbf{x})^{-1} \mathbf{a}_0} \quad (18)$$

and

$$\Psi_{CB}(\mathbf{x}) \approx \frac{4\pi d(\mathbf{x})^2}{\alpha^2 \rho_o c} \cdot \frac{1}{\mathbf{a}_0^T \mathbf{R}_s(\mathbf{x})^{-1} \mathbf{a}_0}, \quad (19)$$

where $d(\mathbf{x})$ is the mean distance from voxel $\mathbf{x}$ to the array elements in Eq. (11).

In the CB, the *assumed* steering vector $\mathbf{a}_0$ for the pre-steered and amplitude-compensated data vector $\mathbf{D}(\mathbf{x}) \mathbf{s}(\mathbf{x}, t)$ is the $\mathbf{1}$ vector. In many cases involving experimental data, the assumed steering vector is not exactly equal to $\mathbf{a}_0 = \mathbf{1}$ due to array imperfections (cross-coupling and variations in element positions), poor array calibration (gain mismatch), and a varying speed of propagation in inhomogeneous media. The CB performance is degraded in this situation as a mismatch exists between the steering vector in the received data and the assumed steering vector used to design the

Capon weight vector in Eqs. (15) and (16). For reasons such as these, Stoica *et al.*¹⁷ developed the RCB that allows the assumed steering vector $\bar{\mathbf{a}} = \mathbf{1}$ to vary by a set parameter ϵ that corresponds to the array calibration uncertainty. We review the RCB algorithm next as a robust version of the standard CB in Eqs. (16) and (17) using the approximation in Eq. (12), while RCB without this approximation is obtained by replacing $\mathbf{R}_s(\mathbf{x})$ with $\mathbf{D}(\mathbf{x}) \mathbf{R}_s(\mathbf{x}) \mathbf{D}(\mathbf{x})$. Later, in Sec. III C, the advantage of the RCB over CB is demonstrated using experimental data.

If we define the nominal steering vector as $\bar{\mathbf{a}} = \mathbf{1}$ for pre-steered data, then the RCB constrained optimization problem can be written as

$$\min_{\mathbf{a}} \mathbf{a}^T \mathbf{R}_s(\mathbf{x})^{-1} \mathbf{a} \text{ subject to } \|\mathbf{a} - \bar{\mathbf{a}}\|^2 \leq \epsilon. \quad (20)$$

The solution for the new steering vector requires the inverse of the covariance matrix

$$\hat{\mathbf{a}}(\mathbf{x}) = \bar{\mathbf{a}} - \mathbf{R}_s^{-1}(\mathbf{x}) \bar{\mathbf{a}}. \quad (21)$$

The inverse covariance matrix can be expressed in terms of the eigenvalue decomposition, and the constrained optimization in Eq. (20) can be expressed as an unconstrained optimization using the method of Lagrange multipliers. If we note

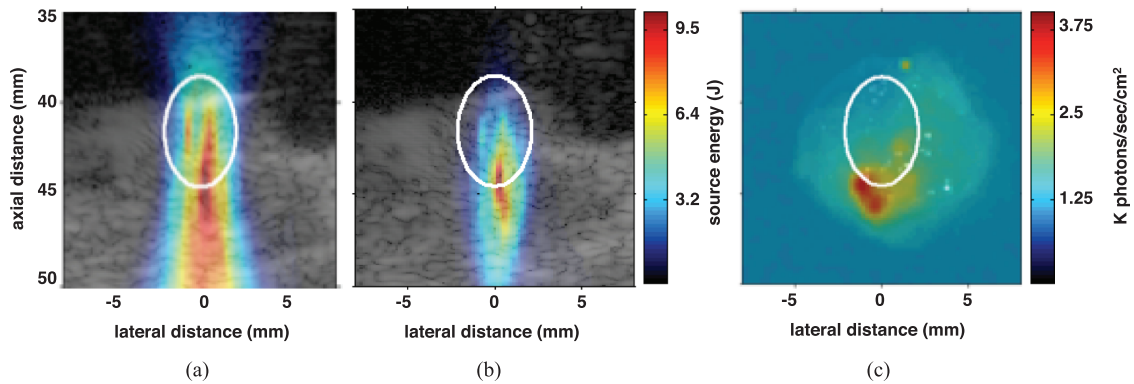


FIG. 3. (Color online) Transaxial PAM images for *in vivo* animal experiments of viral therapy with (a) TEA-PAM and (b) RCB-PAM overlaid on the B-mode images showing the tumors being treated and (c) the luminescence image showing the location of delivered oncolytic virus. Note that the HIFU focus is the ellipse outlined in white.

the eigenvalue decomposition of the covariance matrix as $\mathbf{R}_s(\mathbf{x}) = \mathbf{U}\mathbf{V}\mathbf{U}^T$, then we can rewrite Eq. (21) as

$$\hat{\mathbf{a}}(\mathbf{x}) = \bar{\mathbf{a}} - \mathbf{U}(\mathbf{I} + \lambda\mathbf{V})^{-1}\mathbf{U}^T\bar{\mathbf{a}}, \quad (22)$$

where the Lagrange multiplier is λ and is determined as follows. Tight bounds on λ have been determined,¹⁸ and starting from them, it takes only a few iterations of Newton's method to produce the value of λ as the solution to a nonlinear equation.¹⁸ Additionally, it should be noted that the value of the Lagrange multiplier, λ , depends on ϵ in Eq. (20). The new steering vector estimate is then rescaled to properly adjust for the scaling ambiguity^{17,18} so that

$$\hat{\mathbf{a}}_0(\mathbf{x}) = \frac{\hat{\mathbf{a}}N}{\|\hat{\mathbf{a}}\|}. \quad (23)$$

The RCB formulation¹⁸ allows the computation of power (or energy) without the need for explicit weight computation. Using this result with proper normalization to convert to units of energy, the RCB estimated energy at the voxel $\mathbf{x}$ is

$$\begin{aligned} \Psi_{\text{RCB}}(\mathbf{x}) &= \frac{4\pi d(\mathbf{x})^2}{\alpha^2 \rho_o c} \cdot \frac{1}{\hat{\mathbf{a}}_0(\mathbf{x})\mathbf{U}\mathbf{V}(\lambda^{-2}\mathbf{I} + 2\lambda^{-1}\mathbf{V} + \mathbf{V}^2)^{-1}\mathbf{U}^T\hat{\mathbf{a}}_0(\mathbf{x})}. \end{aligned} \quad (24)$$

As with TEA, the entire map of the acoustic activity is created from the energy $\psi_{\text{RCB}}(\mathbf{x})$ for each voxel position $\mathbf{x} \in B$. One of the great advantages, however, of using the CB- or RCB-based PAM is that the resolution and interference rejection is greatly improved.

C. Simulation: Interference artifact in PAM images

It has been observed that experimental TEA-PAM images are often characterized by a “tail effect” artifact that may be seen in the left panels of Figs. 2 (*in vitro*) and 3 (*in vivo*). These TEA-PAM images show large amplitudes at ranges beyond the actual cavitation region, which is indicated by the ellipse. It is hypothesized that the two main causes of the tail effect in TEA are multiple bubbles and potentially multi-bubble interaction coupled with the beam width of the axial point spread function (PSF) of TEA, which grows quadratically away from the HIFU transducer.¹² By coherently combining all bubble emissions received at the array as coming from a particular voxel, TEA-PAM will be subject to greater interference between individual bubble signals at a particular voxel than the Capon beamformer-passive acoustic mapping (CB-PAM) or RCB-PAM, which are formulated specifically to reject such interference between sources. The PSF of the TEA-PAM algorithm causes blurring of the estimated source field, with increasing amount as one moves farther away from the array due to the reduced axial resolution. For individual point sources or a small number of bubbles, this is not an issue. Considering the effect at the level of an individual voxel to be imaged, multiple bubbles will act as interferers and will increase the axial blurring at larger depths.

One challenge in developing methods for exploring the imaging of multiple bubbles is that there are few multi-bubble models that do not either make simplifying assumptions or become intractable beyond two or three bubbles. For this reason, a statistical signal model based on the superposition of bubble signals received by the array was chosen for simplicity and flexibility. The bubble model chosen is from Vokurka^{29–31} and defines the signals received at a sensor as a random pulse train. The amplitude, phase, and time constant of the pulses are hypothesized as independent, random variables. For multiple bubbles recorded on a single sensor, the model has the form

$$s_j(t) = \sum_{n=-\infty}^{\infty} \sum_{k=1}^K \frac{p_{kn}}{d_j(\mathbf{x}_k)} \exp\left(\frac{-|t - nT_O - \phi_{kn} - \tau_{j,k}|}{\theta_{kn}}\right), \quad (25)$$

where T_O is the pulse period and K is the fixed number of bubbles. In the original model,²⁹ the number of bubbles varies per pulse as a Poisson random variable, but here it is fixed to simplify the model. The phase offset in a pulse period, ϕ_{kn} , the time constant, θ_{kn} , and peak acoustic pressure, p_{kn} , are all assumed to be normally distributed random variables. This model has been shown to create power spectra that agree closely with experimental results^{29,31} of bubble emissions. Propagation from bubble locations to the array was modeled as linear propagation in a homogeneous medium, though layered media would be a useful extension.

To test the above hypothesis, we considered a case in which the model in Eq. (25) is used to simulate a scenario shown in Fig. 1 in water. The nominal distance from the HIFU focus to the PAM array is 40 mm, and the number of bubble sources is increased from $K = 2$ to 100 bubbles, distributed spatially inside the focal area with a normal distribution. The standard deviation of the distribution is 0.5 mm laterally and 1 mm axially, reflective of the $3.6 \text{ mm} \times 5.6 \text{ mm}$ beam width of the array (defined at -6 dB). The HIFU is on for $100 \mu\text{s}$ at $f_0 = 1 \text{ MHz}$. The bubbles are assumed to exist every HIFU cycle with random parameters $p_{kn} \sim \mathcal{N}(3 \text{ MPa}, 10 \text{ kPa})$, $\phi_{kn} \sim \mathcal{N}(1 \mu\text{s}, 14 \text{ ns})$, and $\theta_{kn} \sim \mathcal{N}(2 \text{ ns}, 0.5 \text{ ns})$, where $\mathcal{N}(a, b)$ denotes a normal random variable with mean a and standard deviation b . For each value of K , ten iterations of the simulation were completed with each iteration having randomly placed bubbles. The total integration time for PAM is $100 \mu\text{s}$.

D. In-vitro and in-vivo experimental apparatus

The same experimental configuration is used to test the algorithms both *in vitro* and *in vivo*. The configuration is shown in Fig. 1(a), where two 500 kHz focused, US transducers (Sonic Concepts, Bothell, WA) (outer diameter: 64 mm, radius of curvature: 62.6 mm, focal length: 65 mm; Sonic Concepts, Inc., Bothell, WA) were aligned perpendicularly to each other with their foci intersecting. This formed a $5.4 \times 3.6 \times 1.4 \text{ mm}$ (axial, lateral, and elevation) focal volume measured by the full width at half-maximum of the peak rarefactional pressure (PRP) that was acquired by scanning a $75 \mu\text{m}$ diameter needle hydrophone (Precision

Acoustics Ltd., Dorchester, United Kingdom). The transducers are connected to their custom matching networks and driven by two function generators (model: 33250 A; Agilent, Santa Clara, CA) through two 50 dB power amplifiers (models: A300 and 1140LA; Electronics and Innovation, Rochester, NY).

The passive array system consisting of a Zonare Z.one US imaging engine (Zonare Medical Systems, Mountain View, CA) connected to a Zonare L10-5 probe (center frequency: 7.14 MHz, bandwidth: 5 MHz), shown in detail in Fig. 1(b), is aligned along the axial direction to the focal volume, but between the two transducers. The L10-5 has 128 elements with spacing 0.3 mm and width of 5 mm. Operating passively, the Z.one acquired 64 elements comprising a 38.4 mm aperture (element 2 to 64 and element 65 to 127) simultaneously.^{3,26,32} The Z.one was operated at a frame rate of 400 Hz with each frame of 60 μ s at a sampling frequency of 50 MHz. PAM total integration time is 60 μ s per map.

For the *in vitro* experiments, a phantom is made from 3% agar and has a center channel of approximately 1.6 mm aligned parallel to the passive array and centered with the focal volume of the transducers as shown in Fig. 1(b). Flowing through the channel is a solution of phosphate buffered saline (PBS) containing the contrast agent Sonovue (Bracco Spa., Milan, Italy), with flow of approximately 0.04 mL/s and an expected concentration of 0.3×10^6 to 1.7×10^6 microbubbles/mL.²⁶ The applied drive signal was 25 cycles of a sine wave, with peak negative pressure 300 kPa, which is above the known pressure to cause both stable and inertial cavitation in Sonovue microbubbles, but was experimentally determined to be well below that of the phantom itself and PBS without microbubbles.³³ The value of ϵ used in RCB-PAM for the *in vitro* experiments was initially determined through trial as 8.45, but later optimally found to be 8.39 (see Sec. III C) through a parameter search.

In the *in vivo* experiments, luciferase expressing tumor cells (estrogen receptor positive, ZR75.1 human breast cancer) were subcutaneously injected into three Balb/c SCID female mice, and allowed to form 100–200 mm³ tumors. For the experiment, the mouse body was placed in a degassed water bath at 37°C, and the tumor was aligned to the US focal volume through guidance of the Zonare Z.one operating in B-mode. During each treatment, AdeHE2F adenovirus (25 μ L/5 $\times 10^9$ copies) with 2 μ L of Sonovue was administered via tail cannula 2 times, followed by 4 subsequent injections of 2 μ L of Sonovue, each every 120 s. US is applied in 80 pulses, each 100 ms sine waves at 1.2-MPa PRP with a 3 s pulse repetition interval. To allow for infection and gene delivery, bioluminescence images were acquired 24 h post treatment, with luciferin having been systemically administered to reveal successful gene transfer through luciferase expression. The value of ϵ used in RCB-PAM was 6.

III. RESULTS AND DISCUSSION

A. Analysis: Interference suppression in PAM

For narrowband array processing and far-field sources, it is well-known that optimal beamformers improve the

resolution and suppress interference, through both analysis and experiments.²⁷

However, the assumptions made in previous analyses are not true of PAM within UT due to the inclusion of the pre-steering step and the presence of both harmonic and broadband components in the received data. In this section, we show analytically that PAM using optimal beamforming with wideband data that is pre-steered to the voxel of interest provides interference suppression similar to the far-field, narrow-band case, and we also consider the effect of correlation between sources. Consider the case of $K = 2$ sources, with the sensor data pre-steered to the location of source 1, denoted by $\mathbf{x}_1$. Defining the vector $\mathbf{1} = [1 \cdots 1]^T$, the correlation matrix is derived in the Appendix, and Eq. (A13) in the Appendix specialized to the case of $K = 2$ sources has the form

$$\mathbf{R}_s(\mathbf{x}_1) = R_p(0; 1, 1) \cdot \mathbf{1} \cdot \mathbf{1}^T + (\mathbf{r}_2 \cdot \mathbf{1}^T + \mathbf{1} \cdot \mathbf{r}_2^T) + \mathbf{R}_{2:2} + \sigma^2 \cdot \mathbf{I} \quad (26)$$

$$= [\mathbf{1} \quad \mathbf{r}_2] \begin{bmatrix} R_p(0; 1, 1) & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{r}_2^T \end{bmatrix} + \mathbf{R}_{2:2} + \sigma^2 \cdot \mathbf{I} \quad (27)$$

$$= \mathbf{ZCZ}^T + \mathbf{Q}, \quad (28)$$

where $R_p(0; 1, 1)$ is proportional to the energy of source 1 (which we are trying to estimate), the vector $\mathbf{r}_2$ is the cross-correlation between the signals from source 1 and source 2 after pre-steering, the matrix $\mathbf{R}_{2:2}$ is the spatial autocorrelation matrix of source 2 signals after pre-steering, and σ^2 is the noise energy. The matrices $\mathbf{Z}$, $\mathbf{C}$, and $\mathbf{Q}$ are defined by comparing Eqs. (27) and (28). In particular, the matrix $\mathbf{Q} = \mathbf{R}_{2:2} + \sigma^2 \mathbf{I}$ is positive definite, so its inverse is also positive definite and admits a square-root factorization $\mathbf{Q}^{-1} = \mathbf{Q}^{-T/2} \mathbf{Q}^{-1/2}$.

For simplicity, we consider a comparison between the TEA-based PAM image and the CB-based PAM image (rather than RCB-PAM). Then with $\mathbf{a}_0 = \mathbf{1}$, the Capon-based PAM image at $\mathbf{x}_1$ can be expressed as

$$\begin{aligned} \tilde{\Psi}_{\text{CB}}(\mathbf{x}_1) &= \frac{1}{\mathbf{1}^T \mathbf{R}_s(\mathbf{x}_1)^{-1} \mathbf{1}} \\ &= R_p(0; 1, 1) + \frac{1}{\mathbf{1}^T \mathbf{Q}^{-1} \mathbf{1}} + \frac{2}{\mathbf{1}^T \mathbf{Q}^{-1} \mathbf{1}} \mathbf{1}^T \mathbf{Q}^{-1} \mathbf{r}_2 \\ &\quad - \mathbf{r}_2^T \mathbf{Q}^{-T/2} \left[\mathbf{I} - \frac{\mathbf{Q}^{-1/2} \mathbf{1} \mathbf{1}^T \mathbf{Q}^{-T/2}}{\mathbf{1}^T \mathbf{Q}^{-1} \mathbf{1}} \right] \mathbf{Q}^{-1/2} \mathbf{r}_2. \end{aligned} \quad (29)$$

Similarly, the TEA-based PAM image for this case has the form

$$\begin{aligned} \tilde{\Psi}_{\text{TEA}}(\mathbf{x}_1) &= \frac{\mathbf{1}^T \mathbf{R}_s(\mathbf{x}_1) \mathbf{1}}{(\mathbf{1}^T \mathbf{1})^2} \\ &= R_p(0; 1, 1) + \frac{\mathbf{1}^T \mathbf{Q} \mathbf{1}}{(\mathbf{1}^T \mathbf{1})^2} + \frac{2}{\mathbf{1}^T \mathbf{1}} \mathbf{1}^T \mathbf{r}_2. \end{aligned} \quad (30)$$

The scaling factor $4\pi d(\mathbf{x}_1)^2 / \alpha^2 \rho_o c$ is omitted from Eqs. (29) and (30), hence the tilde over the ψ symbols. Note that both

Eqs. (29) and (30) have similar forms. The first term is the energy of source 1 (the desired quantity), while the remaining terms are interference from source 2 and the noise. The second term in both PAM expressions may be interpreted as the “leakage” from the source 2 autocorrelation matrix and noise via the $\mathbf{Q}$ matrix, while the remaining terms are the interference due to the cross-correlation between the two sources.

First, we consider the case that source 1 and source 2 are completely uncorrelated, so $\mathbf{r}_2 = \mathbf{0}$, and a low-noise environment (so the source 1 and source 2 energy values are much larger than the noise energy σ^2), and we assume the matrix $\mathbf{R}_{2,2}$ has rank D , corresponding to the source 2 signals in the pre-steered data lying in a D -dimensional subspace. Then the eigendecomposition of $\mathbf{Q} = \mathbf{R}_{2,2} + \sigma^2 \mathbf{I}$ is

$$\mathbf{Q} = \sum_{j=1}^D (\lambda_j^2 + \sigma^2) \mathbf{u}_j \mathbf{u}_j^T + \sum_{j=D+1}^N \sigma^2 \mathbf{u}_j \mathbf{u}_j^T, \quad (31)$$

where $\mathbf{u}_j$ are the orthonormal eigenvectors and λ_j^2 are the eigenvalues of $\mathbf{R}_{2,2}$. Using this result, the PAM images for the pre-steered location in Eqs. (29) and (30) with $\mathbf{r}_2 = \mathbf{0}$ are then

$$\begin{aligned} \tilde{\Psi}_{\text{CB}}(\mathbf{x}_1) &= R_p(0; 1, 1) \\ &+ \frac{1}{\sum_{j=1}^D \frac{1}{\lambda_j^2 + \sigma^2} (\mathbf{1}^T \mathbf{u}_j)^2 + \frac{1}{\sigma^2} \sum_{j=D+1}^N (\mathbf{1}^T \mathbf{u}_j)^2}, \end{aligned} \quad (32)$$

$$\tilde{\Psi}_{\text{TEA}}(\mathbf{x}_1) = R_p(0; 1, 1) + \frac{1}{N^2} \sum_{j=1}^D \lambda_j^2 (\mathbf{1}^T \mathbf{u}_j)^2 + \frac{\sigma^2}{N}. \quad (33)$$

The leakage from source 2 can be significant in TEA if the $\mathbf{1}$ vector has nonzero projection on the eigenvectors of $\mathbf{R}_{2,2}$ via the $(\mathbf{1}^T \mathbf{u}_j)$ factors in Eq. (32), and the leakage grows in proportion to the eigenvalues (which are proportional to the energy of source 2). If source 2 is strong, then the CB suppresses the strong interference due to the $1/(\lambda_m^2 + \sigma^2)$ scaling in the denominator of Eq. (32), and the suppression *improves* as the eigenvalues increase. In the case that one of the eigenvectors is proportional to $\mathbf{1}$, say $\mathbf{u}_1 = \mathbf{1}/\sqrt{N}$, then both TEA and Capon produce the estimate $R_p(0; 1, 1) + (\lambda_1^2 + \sigma^2)/N$. This is reasonable since some of the source 2 energy is aligned with the steering vector for source 1 and is therefore passed by the CB, while the remainder of the source 2 energy is orthogonal to $\mathbf{1}$ and is suppressed by the TEA and CBs.

This analysis partially explains the appearance of the tail artifact in the TEA-PAM images, though additional interference will be present due to the third term in Eq. (30) when $\mathbf{r}_2 \neq \mathbf{0}$ (corresponding to correlation between the source 1 and source 2 signals in the pre-steered data). However, this correlation will also introduce bias in the

Capon (and RCB) estimates, similar to the well-known phenomenon in Capon-based beamformers for narrowband, far-field scenarios.³⁴ When $\mathbf{r}_2 \neq \mathbf{0}$, the CB estimate in Eq. (29) contains a third and fourth term that generally cancels part of the source 1 energy due to the correlation with source 2, producing a negative bias (i.e., decrease) in the estimate of source 1 energy. The third terms in Eqs. (29) and (30) may be positive or negative, while the fourth term in Eq. (29) is either zero or it decreases the estimate, corresponding to partial cancellation of the energy of source 1 by the CB. However, in the UT application of interest in this work, the cavitation bubble signals are brief in time so that when signals from different cavitation bubbles arrive at the array, they are usually uncorrelated, so signal cancellation does not appear to be a significant problem in PAM, as will be evidenced by the experimental results.

B. Simulation: Tail artifact reduction

Moving from the analysis to simulation, the goal was to show that increasing numbers of bubbles within the imaging region leads to a tail artifact that is exhibited in TEA, but also that an optimal beamforming algorithm can reduce this artifact. The resulting images for both TEA-PAM and RCB-PAM are shown in Fig. 4. As can be easily noticed, TEA-PAM exhibits the tail artifact behavior, namely, the appearance of an elongated region of acoustic energy beyond the HIFU “focal region.” As the positions of the bubbles are totally within the focal region by the simulation parameters, no true sources exist in the area beyond the focal region. This means that any perceived post-focal energy is solely due to the TEA algorithm artifact. These results support the hypothesis that the effect seen in TEA-PAM images of experimental data is an artifact of the bubbles interfering with each other rather than being due to actual bubbles that are present behind the HIFU focal region.

To quantify the degree to which the RCB algorithm suppresses this tail artifact behavior and maintains acceptable estimation of bubble energy at the focus, the following two metrics are proposed, the Suppression Ratio and Energy Estimation Ratio. The ratios are based on the maximum voxel energy in the PAM image for two distinct regions, that in the elliptical focal area and that in the post-focal area (i.e., outside of the elliptical focal area but at greater axial distance), and are defined as

$$\text{Suppression Ratio} = \frac{\max_{\mathbf{x} \in \{\text{post-focus}\}} \Psi_{\text{RCB}}(\mathbf{x})}{\max_{\mathbf{x} \in \{\text{focus}\}} \Psi_{\text{RCB}}(\mathbf{x})}, \quad (34)$$

$$\text{Energy Estimation Ratio} = \frac{\max_{\mathbf{x} \in \{\text{focus}\}} \Psi_{\text{RCB}}(\mathbf{x})}{\max_{\mathbf{x} \in \{\text{focus}\}} \Psi_{\text{TEA}}(\mathbf{x})}. \quad (35)$$

The Suppression Ratio measures how RCB reduces the tail artifact compared to the focal energy; and if we assume that the peak energy of TEA is a comparable measure of the

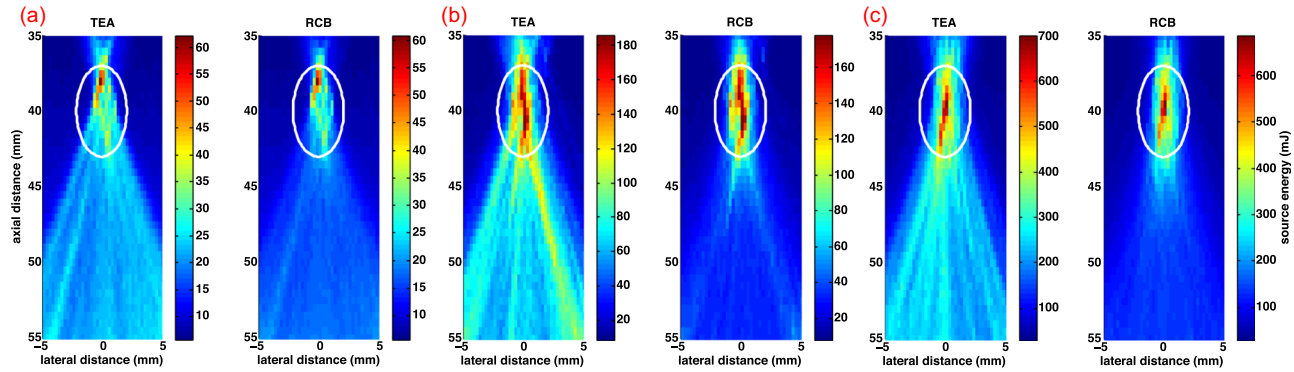


FIG. 4. (Color online) PAM images for simulated data for increasing numbers of sources: (a) 9 sources, (b) 40 sources, and (c) 100 sources. For RCB, $\epsilon = 8.45$. Note that the HIFU focus is the ellipse outlined in white.

true bubble energy (which is unknown), then the Energy Estimation Ratio measures how well RCB performs in preserving this estimate. In the case of many bubbles, it is acknowledged however that the peak energy of TEA may not be the best estimate of true bubble energy due to biasing. Ideally, the Suppression Ratio should be small, while the Energy Estimation Ratio should be close to unity.

The simulated data from the signal model was processed using TEA-PAM and RCB-PAM to determine first, if the tail artifact can be created with increasing number of bubbles, and second, how the algorithms compare with regards to suppressing (or not suppressing) the artifact and estimating the bubble energy. The results for increasing number of bubbles are shown in Fig. 5, plotting mean and the standard errors of the Suppression Ratio and Energy Estimation Ratio for the ten trials. Three representative PAM images are shown in Fig. 4 for 9, 40, and 100 bubbles. Note in Fig. 4 the strong presence of the tail artifact that extends from the HIFU focal region to the furthest axial distance in TEA-PAM. This artifact is greatly reduced in RCB-PAM, where the acoustic activity is confined mainly to the HIFU focal region. Note that the value of ϵ parameter chosen through trial to be 8.45, though it was not strongly dependent on the parameter choice. The Suppression Ratio in Fig. 5

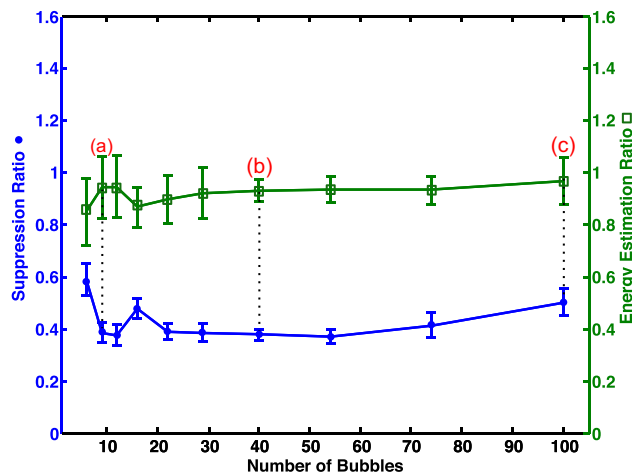


FIG. 5. (Color online) Suppression Ratio and Energy Estimation Ratio for increasing number of simulated bubble sources where: (a) 9 sources, (b) 40 sources, and (c) 100 sources with 10 trials at each data point.

demonstrates this as well, with RCB-PAM removing consistently 40% to 50% of the tail artifact in spite of increasing bubbles. Note also that most of the remaining artifact in RCB-PAM is located in the first 2 mm or so beyond the HIFU focus, while the rest of the post-focal region is closer to the level of background noise. In terms of the Energy Estimation Ratio, RCB-PAM preserves the TEA-PAM energy estimate at over 90% for all the cases tested.

C. Experimental: *In vitro* model

As described in Sec. II D, the cavitation nuclei in the form of Sonovue microbubbles exist only in the flow channel, and the HIFU transducer focus (outlined in the white ellipse in Fig. 2) overlaps the channel in only one area. It is expected that the PAM image should only exhibit cavitation activity confined to the intersection of the two. The TEA-PAM image overlaid on B-mode images of the phantom for this experiment is shown in the left panel of Fig. 2, where US arrives from the top of the image. One can also see the front (axial distance = 27 mm) and the back (axial distance = 44 mm) of the phantom, as well as the approximately 1.6 mm channel (axial distance = 37 mm). The TEA-PAM image suggests that much of the cavitation takes place outside of the flow channel and axially away from the flow channel, up to 1 cm away, which does not agree with the expected cavitation in the experimental setup. In Fig. 2, the same data is processed using CB-PAM in the center panel, which shows only activity post-focally at very high levels. The CB-PAM is completely inaccurate for this data set due to a poor beamforming weight vector, which is explained in Sec. III D. Conversely, the RCB-PAM image for ($\epsilon = 8.45$) is shown in Fig. 2 in the right panel and shows bubble activity of approximately the same extent as the actual (expected) cavitation region of the focus. There does exist some low level activity at the rear boundary of the phantom, which may be due to low levels of bubble activity at the rear face, reflections of the inertial cavitation emissions, or residual reflection of the incident wave at higher harmonics of the fundamental within the bandwidth of the L10-5.

While the peak negative pressure used in the experiment could induce sonoporation,³⁵ it is not expected that enough mechanical disruption of the channel has occurred to transport microbubbles up to 1 cm into the phantom to cause

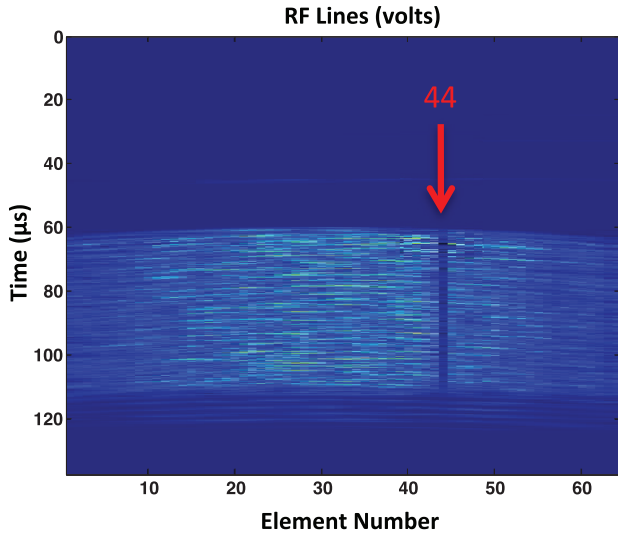


FIG. 6. (Color online) *In vitro* experimental radio frequency data for a single frame showing the variance of sensitivity on the Zonare L10-5, with the highlighted weak element 44.

cavitation³⁶ as seen in the TEA-PAM images. Nor is it expected that endogenously-nucleated cavitation is occurring in the phantom at 300 kPa, as this is well below the cavitation threshold of the phantom. Instead, the setup favors the conclusion that the post-focal heavy tail is an imaging artifact caused by TEA-PAM coherently combining multiple cavitation events, as this effect is similarly evidenced in the simulation examples. Further, in this experiment it was known that for the sample of SonoVue provided, there were potentially thousands of nuclei in the volume intersected by the US focal volume and flow channel (2000–12 000 microbubbles for the expected concentrations).

Interestingly, the highly inaccurate CB-PAM result can be attributed to steering vector mismatch (specifically, gain mismatch) due to element 44 having poor sensitivity compared to the other elements, as shown in Fig. 6 (highlighted by the arrow). The condition number of the covariance matrix is too large, and thus the CB-PAM weight vector cannot compensate for this mismatch. This shows the real advantage

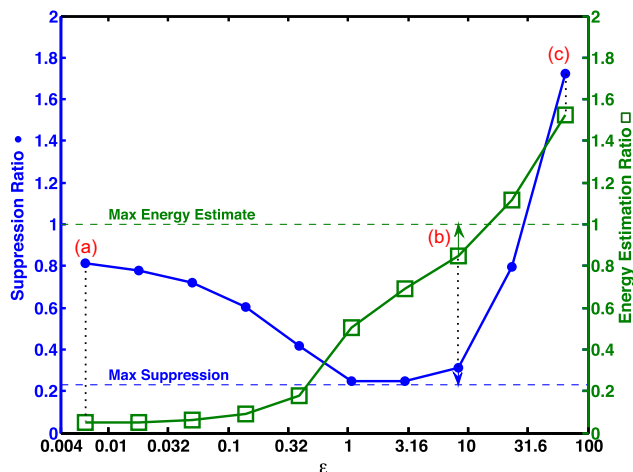


FIG. 7. (Color online) Suppression ratio and energy estimation ratio for *in vitro* experimental RCB-PAM images for increasing levels of the parameter.

of the robustness of RCB-PAM in experiments using real arrays in that the resulting image is less sensitive to array calibration and sensitivity. As is well known, diagonal loading can also compensate for this in CB, but the RCB has this capability built-in and automatically determines the proper loading level based on the data.¹⁸

The optimal value of ϵ will depend on the data presented to the RCB, but in general a larger value will allow the algorithm to suppress more interference. To exclude the trivial solution, it is assumed¹⁸ that $\|\bar{\mathbf{a}}\|^2 > \epsilon$. With the constraint in Eq. (20), this assumption provides some guidance in the range of values that ϵ can take. Practically speaking, this will be the range of $\epsilon = \{N/\gamma, N\}$, where γ is some large number (e.g., 10 000). To test this, the previous data set was re-run using RCB-PAM with varying ϵ parameter and the Suppression Ratio and Energy Estimation Ratio computed, with the results displayed in Fig. 7. Representative RCB-PAM images for $\epsilon = 0.05, 8.39, 64$ are shown in Fig. 8. From the ratios in Fig. 7, it can be observed that there is a trade-off between maximal suppression of interference and maximal estimation of energy with the ideal value being around $\epsilon = 8.39$. The RCB-PAM images in Fig. 8 confirm the general trend that small values ($\epsilon = 0.05$, left) do not completely suppress the interference relative to the focus and have a diminished energy estimate. Conversely, too high a value of ϵ completely accentuates the interference artifact, and the energy at the focus is elevated.

D. Experimental: *In vivo* model

PAM was used to monitor US-enhanced viral therapy in an *in vivo* animal model as described in Sec. II D. As the key issue in treating solid tumors with oncolytic adenoviruses is getting the therapeutic agent into the tumor itself to start self-amplifying, focused US and cavitation nucleated from systemically injected microbubbles will aid in this transport.

The TEA-PAM image was overlaid on the B-mode image and shown in Fig. 3(a). As with simulation and *in vitro* experiments, TEA-PAM displayed a significant tail artifact of acoustic activity in the post-focal region. While some cavitation activity at the back of the focus is not unlikely owing to the presence of more potential nuclei in the healthy vasculature surrounding the tumor, the HIFU focus was aligned directly on the tumor, with the rear of the focus at the back extent of the tumor to limit any potential damage to the healthy vasculature. By comparison though, the RCB-PAM ($\epsilon = 6$) shows much of the acoustic activity occurs in the focal region with greatly reduced post-focal activity. The key corroboratory evidence is the bioluminescence image shown in Fig. 3(c). This shows that luciferase expression was high at the focal region boundary and correlates very well with the RCB-PAM image of acoustic activity for the treatment. While there is an offset in the lateral distance, this is of course to be expected, as the RCB-PAM maps show the location of cavitation activity leading to release, while drugs will diffuse in biological tissue following release. The key point is that the RCB-PAM cavitation maps correctly predict the location and approximate spatial

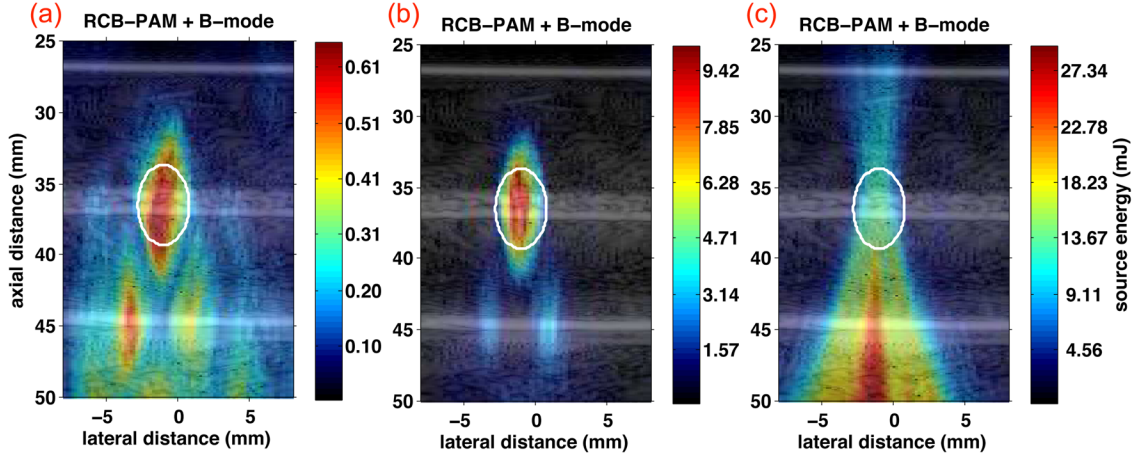


FIG. 8. (Color online) *In vitro* experimental RCB-PAM images for increasing levels of the parameter ϵ : (a) $\epsilon = 0.05$, (b) $\epsilon = 8.39$, and (c) $\epsilon = 64$.

spread of release. Further *in vivo* results from this study will be discussed in future published work.

IV. CONCLUSIONS

An improved method for near-field, wideband PAM has been introduced that takes advantage of optimal beamforming. CB-PAM and RCB-PAM were introduced, with RCB-PAM being preferred in practice for its ability to handle steering vector mismatch and array calibration issues, while Capon is used to analyze interference suppression and source correlation effects. Both issues are of key interest in PAM as all imaging media will be inhomogeneous and thus have multiple propagation velocities, and because PAM applications will generally employ “black-box” diagnostic US probes that can be difficult to calibrate for passive imaging. For conventional TEA-PAM algorithms, a “tail” artifact was observed at modest pressures. The artifact was explained through a simulated bubble model that showed the same effect was caused by multi-bubble interference. This was supported by analytical results showing that multiple bubbles create interference that the TEA-PAM method cannot generally reduce, but which RCB-PAM can reduce. Finally, experimental results were presented that compared the original TEA-PAM with the improved RCB-PAM. It was shown that acoustic region of activity in RCB-PAM more accurately corresponded to the region predicted by the HIFU focus and the experimental setup (i.e., the flow channel with cavitation producing microbubbles). Similarly, in *in vivo* experiments, the acoustic region of activity imaged by RCB-PAM better corresponded to the maximum area of luciferase expression in luminescence imagery. Thus RCB-PAM has been demonstrated as a marked improvement in the development of an US monitoring system by more reliably imaging the extent of the acoustic activity during treatment.

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APPENDIX: CORRELATION MATRIX DERIVATION

In this appendix, we derive the structure of the correlation matrix in Eq. (7) for the signal model for the pre-steered data in Eq. (5). Let us begin by approximating the distances $d_j(\mathbf{x}_k)$ from source k to sensor j in Eq. (5) by the mean distance from source k to all the sensors,

$$d(\mathbf{x}_k) = \frac{1}{N} \sum_{j=1}^N d_j(\mathbf{x}_k), \quad (\text{A1})$$

so the pre-steered data in Eq. (5) is well-approximated by

$$s_j(t + \tau_j(\mathbf{x})) \approx \sum_{k=1}^K \frac{1}{d(\mathbf{x}_k)} p_k(t + \tau_j(\mathbf{x}) - \tau_{j,k}) + v_j(t + \tau_j(\mathbf{x})), \quad j = 1, \dots, N. \quad (\text{A2})$$

We define the cross-correlation between the scaled signals from sources k and l with lag τ as

$$R_p(\tau; k, l) = \frac{1}{d(\mathbf{x}_k)d(\mathbf{x}_l)} \int_{t_0}^{t_0+T} p_k(t + \tau) p_l(t) dt, \quad (\text{A3})$$

where we assume that the time interval of length T includes the relevant cavitation activity that is being monitored. Note that $R_p(0; k, k)$ is proportional to the energy of source k . Then the correlation matrix $\mathbf{R}_s(\mathbf{x})$ of pre-steered sensor data is an $N \times N$ matrix with elements

$$\begin{aligned} [\mathbf{R}_s(\mathbf{x})]_{m,n} &= \int_{t_0}^{t_0+T} s_m(t + \tau_m(\mathbf{x})) s_n(t + \tau_n(\mathbf{x})) dt \\ &= \sum_{k=1}^K \sum_{l=1}^K R_p(\tau_m(\mathbf{x}) - \tau_{m,k} \\ &\quad - (\tau_n(\mathbf{x}) - \tau_{n,l}); k, l) + \sigma^2 \delta_{m,n}, \end{aligned} \quad (\text{A4}) \quad (\text{A5})$$

where σ^2 is the noise energy in the observation interval. (We have assumed for analytical simplicity that the noise is uncorrelated at distinct sensors and has the same energy at each sensor. This assumption becomes more accurate as the observation interval T increases.) Many of the cross-terms

for $k \neq l$ in Eq. (A5) may equal zero if the source signals are trains of brief, impulsive waveforms as in the Vokurka model [Eq. (25)]. If the signals from distinct sources are uncorrelated in the pre-steered data, then Eq. (A5) reduces to

$$[\mathbf{R}_s(\mathbf{x})]_{m,n} = \sum_{k=1}^K R_p(\tau_m(\mathbf{x}) - \tau_{m,k} - (\tau_n(\mathbf{x}) - \tau_{n,k}); k, k) + \sigma^2 \delta_{m,n}. \quad (\text{A6})$$

Suppose that the imaging position $\mathbf{x}$ is equal to $\mathbf{x}_1$, the position of source 1. Then for each sensor, the pre-steering delays are $\tau_m(\mathbf{x}) = \tau_{m,1}$, and the correlation matrix elements in (A5) have the structure

$$[\mathbf{R}_s(\mathbf{x}_1)]_{m,n} = R_p(0; 1, 1) \quad (\text{A7})$$

$$+ \sum_{k=2}^K R_p(\tau_{m,1} - \tau_{m,k}; k, 1) + \sum_{l=2}^K R_p(-(\tau_{n,1} - \tau_{n,l}); 1, l) \quad (\text{A8})$$

$$+ \sum_{k=2}^K \sum_{l=2}^K R_p(\tau_{m,1} - \tau_{m,k} - (\tau_{n,1} - \tau_{n,l}); k, l) \quad (\text{A9})$$

$$+ \sigma^2 \delta_{m,n}. \quad (\text{A10})$$

Note that $R_p(\tau; k, l) = R_p(-\tau; l, k)$, and define the vectors $\mathbf{r}_2, \dots, \mathbf{r}_K$ to collect the elements in (A8),

$$\mathbf{r}_k = \begin{bmatrix} R_p(\tau_{1,1} - \tau_{1,k}; k, 1) \\ \vdots \\ R_p(\tau_{M,1} - \tau_{M,k}; k, 1) \end{bmatrix}, \quad k = 2, \dots, K, \quad (\text{A11})$$

and define the matrix $\mathbf{R}_{2:K}$ with elements given by (A9),

$$[\mathbf{R}_{2:K}]_{m,n} = \sum_{k=2}^K \sum_{l=2}^K R_p(\tau_{m,1} - \tau_{m,k} - (\tau_{n,1} - \tau_{n,l}); k, l). \quad (\text{A12})$$

Then the correlation matrix $\mathbf{R}_s(\mathbf{x}_1)$ of sensor data pre-steered to the location of source 1 has the structure

$$\mathbf{R}_s(\mathbf{x}_1) = R_p(0; 1, 1) \cdot \mathbf{1} \cdot \mathbf{1}^T + \sum_{k=2}^K (\mathbf{r}_k \cdot \mathbf{1}^T + \mathbf{1} \cdot \mathbf{r}_k^T) + \mathbf{R}_{2:K} + \sigma^2 \cdot \mathbf{I}, \quad (\text{A13})$$

where the steering vector for source 1 is clearly $\mathbf{1}$ in the first term. The second term in Eq. (A13) is a summation that corresponds to correlations between source 1 and the other sources, and the third and fourth terms are the spatial autocorrelation matrix of the signals from sources $2, \dots, K$ in the pre-steered data and the noise, respectively.

- ¹C.-C. Coussios and R. Roy, "Applications of acoustics and cavitation to noninvasive therapy and drug delivery," *Annu. Rev. Fluid. Mech.* **40**, 395–420 (2008).
- ²M. Gyöngy and C.-C. Coussios, "Passive spatial mapping of inertial cavitation during HIFU exposure," *IEEE Trans. Biomed. Eng.* **57**(1), 48–56 (2010).
- ³C. Jensen, R. Ritchie, M. Gyöngy, J. Collin, T. Leslie, and C.-C. Coussios, "Spatiotemporal monitoring of high-intensity focused ultrasound therapy passive acoustic mapping," *Radiology* **262**(1), 252–261 (2012).
- ⁴S. Mo and C.-C. Coussios, "Ultrasound-enhanced drug delivery for cancer," *Expert Opin. Drug Delivery* **9**(12), 1525–1538 (2012).
- ⁵M. Bazan-Peregrino, B. Rifai, R. Carlisle, J. Choi, C. Arvanitis, L. Seymour, and C.-C. Coussios, "Cavitation-enhanced delivery of a conditionally replicating oncolytic adenovirus by therapeutic ultrasound," *J. Controlled Release* **169**, 40–47 (2013).
- ⁶C. Arvanitis, M. Livingstone, and N. McDannold, "Combined ultrasound and MR imaging to guide focused ultrasound therapies in the brain," *Phys. Med. Biol.* **58**(14), 4749–4761 (2013).
- ⁷R. Jones, M. O'Reilly, and K. Hynynen, "Transcranial passive acoustic mapping with hemispherical sparse arrays using ct-based skull-specific aberration corrections: A simulation study," *Phys. Med. Biol.* **58**(14), 4981–5005 (2013).
- ⁸V. Salgaonkar, S. Datta, C. Holland, and T. Mast, "Passive cavitation imaging with ultrasound arrays," *J. Acoust. Soc. Am.* **126**(6), 3071–3083 (2009).
- ⁹K. Haworth, T. Mast, K. Radhakrishnan, M. Burgess, J. Kopechek, S. Huang, D. McPherson, and C. Holland, "Passive imaging with pulsed ultrasound insonations," *J. Acoust. Soc. Am.* **132**(1), 544–553 (2012).
- ¹⁰C. Coviello, R. Kozick, A. Hurrell, P. Smith, and C.-C. Coussios, "Thin-film sparse boundary array design for passive acoustic mapping during ultrasound therapy," *IEEE Trans. Ultrason., Ferroelectr., Freq. Control* **59**(10), 2322–2330 (2012).
- ¹¹S. Norton and I. Won, "Time exposure acoustics," *IEEE Trans. Geosci. Remote Sens.* **38**(3), 1337–1343 (2000).
- ¹²M. Gyöngy, "Passive cavitation mapping for monitoring ultrasound therapy," Ph.D. dissertation, University of Oxford, Oxford, United Kingdom, 2011.
- ¹³J. Capon, "High-resolution frequency-wavenumber spectrum analysis," *Proc. IEEE* **57**(8), 1408–1418 (1969).
- ¹⁴J.-F. Synnevåg, A. Austeng, and S. Holm, "Adaptive beamforming applied to medical ultrasound imaging," *IEEE Trans. Ultrason., Ferroelectr., Freq. Control* **54**(8), 1606–1613 (2007).
- ¹⁵J.-F. Synnevåg, A. Austeng, and S. Holm, "Benefits of high-resolution beamforming in medical ultrasound imaging," *IEEE Trans. Ultrason., Ferroelectr., Freq. Control* **56**(9), 1868–1879 (2009).
- ¹⁶I. Holfort, F. Gran, and J. Jensen, "Broadband minimum variance beamforming for ultrasound imaging," *IEEE Trans. Ultrason., Ferroelectr., Freq. Control* **56**(2), 314–325 (2009).
- ¹⁷P. Stoica, Z. Wang, and J. Li, "Robust capon beamforming," *IEEE Signal Proc. Lett.* **10**(6), 172–175 (2003).
- ¹⁸J. Li, P. Stoica, and Z. Wang, "On robust capon beamforming and diagonal loading," *IEEE Trans. Signal Proc.* **51**(7), 1702–1715 (2003).
- ¹⁹Z. Wang, J. Li, and R. Wu, "Time-delay- and time-reversal-based robust capon beamformers for ultrasound imaging," *IEEE Trans. Med. Imag.* **24**(10), 1308–1322 (2005).
- ²⁰B. Guo, J. Li, H. Zmuda, and M. Sheplak, "Multifrequency microwave-induced thermal 1 acoustic imaging for breast cancer detection," *IEEE Trans. Biomed. Eng.* **54**(11), 2000–2010 (2007).
- ²¹Y. Xie, B. Guo, L. Xu, J. Li, and P. Stoica, "Multistatic adaptive microwave imaging for early breast cancer detection," *IEEE Trans. Biomed. Eng.* **53**(8), 1647–1657 (2006).
- ²²The term "passive" is used because, unlike pulse-echo methods, the technique makes no use of any prior information relating to the timing or amplitude of the transmitted pulse in order to detect or localize sound sources (Refs. 23 and 24).
- ²³S. I. Madanshetty, R. A. Roy, and R. E. Apfel, "Acoustic microcavitation: Its active and passive acoustic detection," *J. Acoust. Soc. Am.* **90**(3), 1515–1526 (1991).
- ²⁴R. O. Cleveland, O. A. Sapozhnikov, M. R. Bailey, and L. A. Crum, "A dual passive cavitation detector for localized detection of lithotripsy-induced cavitation in vitro," *J. Acoust. Soc. Am.* **107**, 1745–1758 (2000).

- ²⁵In almost all practical cases, this would be in the near-field of the diagnostic array owing to frequency, speed of sound, and aperture size.
- ²⁶J. Choi and C.-C. Coussios, "Spatio-temporal evolution of cavitation dynamics exhibited by flowing microbubbles during therapeutic ultrasound exposure," *J. Acoust. Soc. Am.* **132**(5), 3538–3549 (2012).
- ²⁷B. D. Van Veen and K. M. Buckley, "Beamforming: A versatile approach to spatial filtering," *ASSP Mag. IEEE* **5**(2), 4–24 (1998).
- ²⁸H. L. Van Trees, *Optimum Array Processing (Detection, Estimation, and Modulation Theory, Part IV)* (Wiley-Interscience, New York, 2002).
- ²⁹K. Vokurka, "Cavitation noise modeling and analyzing," in *CD-ROM Proceedings of Forum Acousticum 2002*, Sevilla, Spain (Sociedad Espanola de Acústica, 2002).
- ³⁰K. Vokurka, "A method for evaluating experimental data in bubble dynamics studies," *Czech. J. Phys. B* **36**, 600–615 (1986).
- ³¹K. Vokurka, "Experimental study of the bubble pulse," *Acustica* **66**, 174–176 (1988).
- ³²M. Gyöngy and C.-C. Coussios, "Passive cavitation mapping for localization and tracking of bubble dynamics," *J. Acoust. Soc. Am.* **128**(4), EL175–EL180 (2010).
- ³³C. Arvanitis, M. Bazan-Peregrino, B. Rifai, L. Seymour, and C.-C. Coussios, "Cavitation-enhanced extravasation for drug delivery," *Ultrasound Med. Biol.* **37**(11), 1838–1852 (2011).
- ³⁴V. Reddy, A. Paulraj, and T. Kailath, "Performance analysis of the optimum beamformer in the presence of correlated sources and its behavior under spatial smoothing," *IEEE Trans. Acoust., Speech, Signal Process.* **35**(7), 927–936 (1987).
- ³⁵H.-D. Liang, J. Tang, and M. Halliwell, "Sonoporation, drug delivery, and gene therapy," *Proc. IMechE Part H: J. Eng. in Med.* **224**, 343–361 (2010).
- ³⁶B. Rifai, C. Arvanitis, M. Bazan-Peregrino, and C.-C. Coussios, "Cavitation-enhanced delivery of macromolecules into an obstructed vessel," *J. Acoust. Soc. Am.* **128**(5), EL310–EL315 (2010).